UAV Satria Talon — Spesifikasi Lengkap

Airfoil: NACA 2412 · **Semi-length:** 2.13 ft · **Root:** 0.78 ft · **Tip:** 0.60 ft · **Sweep:** 0°
Empty: 3.00 lb (1.361 kg) · **MTOW:** 5.25 lb (2.381 kg) · **Wingspan:** 129.8 cm
CG: fwd 0.91 ft · nominal 0.92 ft · aft 0.96 ft

Tentang LE dan MAC

LE (Leading Edge) = tepi depan sayap — titik referensi pengukuran CG secara fisik.

MAC (Mean Aerodynamic Chord) = chord rata-rata aerodinamis yang mewakili seluruh sayap. Satria Talon:
- MAC berada di **31.0 cm dari CL** (kiri dan kanan), di **47.8% semi-span**
- Panjang MAC = **21.03 cm**
- Karena sweep **0°**, LE lurus sejajar — ukur langsung dari tepi depan sayap

1. Airfoil NACA 2412

Karakteristik Airfoil NACA 2412

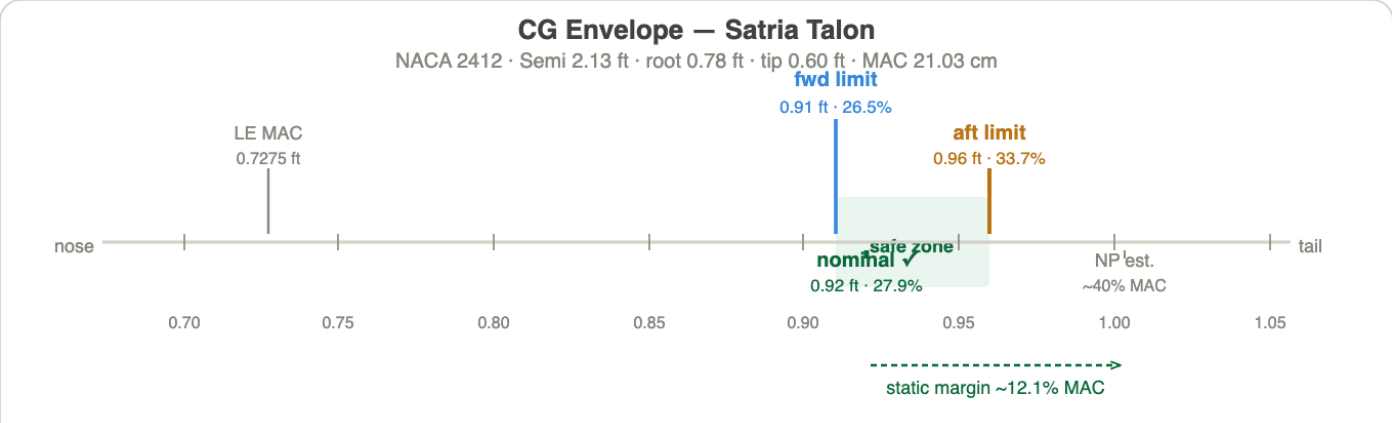
Profil	**NACA 2412**	Camber 2% · max camber 40% chord · t/c 12%
CLmax (Re~700k)	**1.50**	vs generic ~1.30 (+15%)
CD0 total	**0.026**	Lebih rendah dari foil tebal
Oswald efficiency e	**0.82**	Span efficiency baik
L/D max	**12.37**	vs generic ~10.2 (+21%)
Zero-lift AoA	**−2.1°**	Positive camber → slight nose-down at zero lift
Stall AoA	**~16.5°**	Gradual, forgiving — recovery mudah
Cm_ac	**−0.05**	Nose-down pitching moment → CG tidak terlalu belakang
AoA @ cruise	**3.76°**	Dari zero-lift + CL_cruise/Cl_alpha

2. Weight & Balance

Weight & Balance — Satria Talon

Empty weight	3.00 lb	1.361 kg
Maximum weight (MTOW)	5.25 lb	2.381 kg
Payload total	2.25 lb	1.021 kg
LE MAC dari origin	0.7275 ft	22.17 cm
CG fwd limit	0.91 ft	26.5% MAC · 5.56 cm dari LE MAC
CG nominal ✓	0.92 ft	27.9% MAC · 5.87 cm dari LE MAC
CG aft limit	0.96 ft	33.7% MAC · 7.09 cm dari LE MAC (SM 6.3%)
SM @ nominal	–	~12.1% MAC (NP est. ~40%) — AMAN ✓

CG Envelope



Titik CG	Nilai	% MAC	Dari LE MAC	Static Margin	Status
LE MAC (ref)	0.7275 ft	0%	0 cm	—	—
CG fwd	**0.91 ft**	**26.5%**	**5.56 cm**	13.6%	AMAN
CG nominal ✓	**0.92 ft**	**27.9%**	**5.87 cm**	12.1%	**AMAN**
CG aft limit	0.96 ft	33.7%	7.09 cm	6.3%	AMAN
NP (estimasi)	~1.0035 ft	~40%	~8.41 cm	—	ref

Payload Breakdown

Komponen	Berat	Sisa
Empty weight	1.361 kg	1.021 kg
+ Baterai 4S 6000 mAh	+350 g	**671 g**
+ Baterai 4S 8000 mAh	**+450 g**	**571 g**

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3. Geometri Sayap

Wing 1

Parameter	Imperial	Metrik
Semi-length	**2.13 ft**	**64.92 cm**
Wingspan	**4.26 ft**	**129.84 cm**
Root chord	**0.78 ft**	**23.77 cm**
Tip chord	**0.60 ft**	**18.29 cm**
Taper ratio	—	0.7692
Sweep	0.0°	—
Dihedral	0.0°	—
MAC	**0.6900 ft**	**21.03 cm**
Wing area	**2.939 ft²**	**0.2731 m²**
Aspect ratio	—	**6.1739**
Long arm (aero ref)	0.90 ft	27.43 cm
Lat arm	0.19 ft	5.79 cm
Vert arm	0.25 ft	7.62 cm
Aileron	element 4–8 (outboard ~50% span)	—

Wing 2 — Winglet / Strake

Parameter	Nilai
Semi-length	0.43 ft (13.1 cm)
Root chord	0.60 ft (18.3 cm)
Tip chord	0.08 ft (2.4 cm)
Sweep / Dihedral	80° / 20°
Lat arm	2.13 ft (= Wing 1 semi)

Vertical Stabilizer — V-tail (×2)

Parameter	Nilai
Height	0.65 ft (19.8 cm)
Root / Tip chord	0.56 ft / 0.25 ft
Sweep / Dihedral	40° / 45°
Long arm	2.00 ft (60.96 cm)
Elevator + Rudder	aktif el 2–6 → ruddervator

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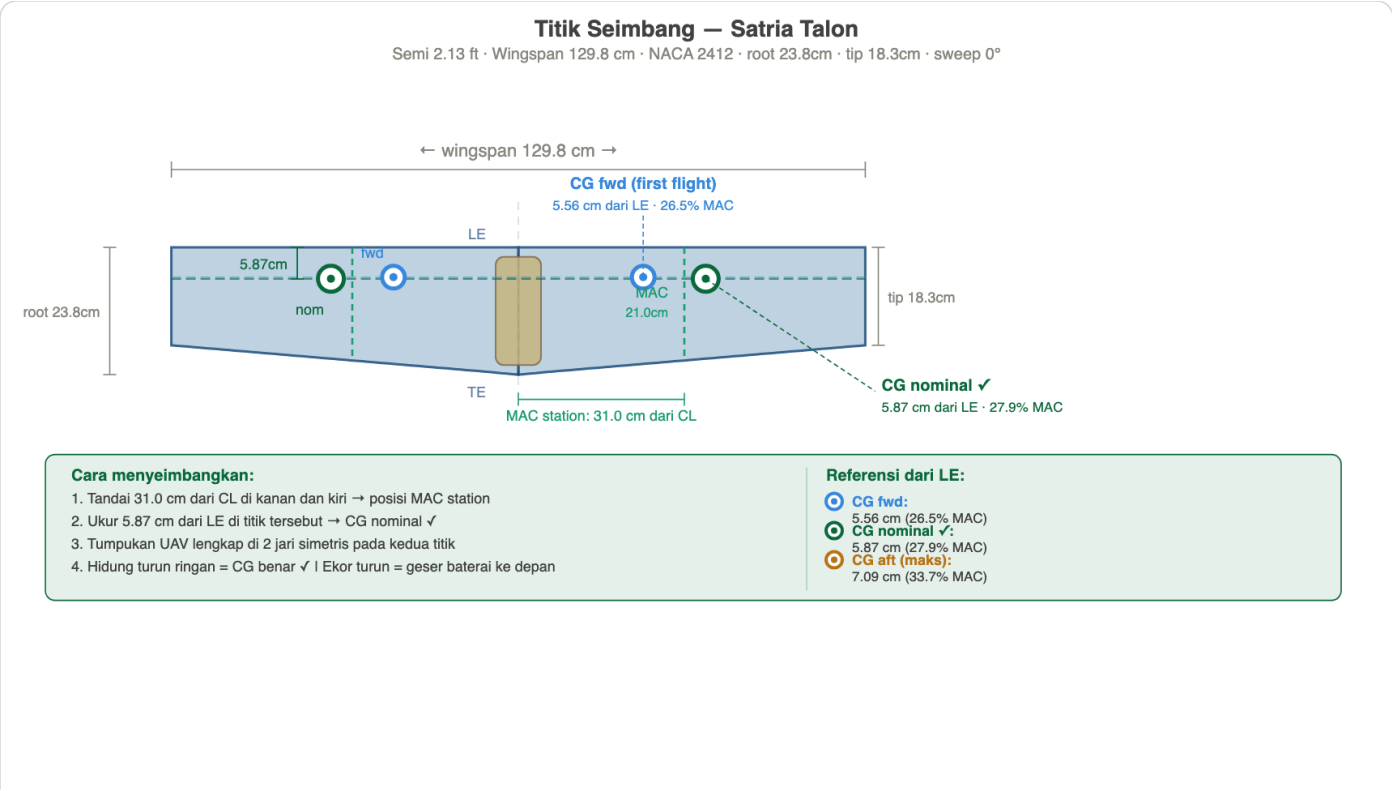
4. Titik Seimbang



Cara Mengukur

1. Tandai **31.0 cm dari CL** di kanan dan kiri → posisi MAC station
2. Di titik tersebut, ukur dari **LE (tepi depan sayap)**:
3. **5.56 cm** → CG fwd (first flight)
4. **5.87 cm** → CG nominal ✓
5. **7.09 cm** → batas maksimum
6. Tumpukan UAV di 2 jari simetris → hidung turun ringan = benar ✓

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5. Performa Aerodinamis

Parameter	Nilai
Wing loading	**85.5 N/m²**
Stall speed Vs	**9.649 m/s (34.7 km/h)**
Min safe airspeed	11.58 m/s (41.7 km/h)
Cruise speed	**14.738 m/s (53.1 km/h)**
AoA cruise	3.76°
VNE	33.77 m/s (121.6 km/h)
L/D max	**12.366**
Cruise power	45.9 W (0.0616 HP)
Peak power	174 W (0.234 HP)
Motor rating min	209 W (0.280 HP)

Mode High Speed

Parameter	80 km/h	100 km/h
Kecepatan	22.2 m/s	**27.8 m/s**
L/D	9.12	6.44
Cruise power	94.0 W (0.126 HP)	**153 W (0.205 HP)**
Peak power	117.5 W (0.158 HP)	191 W (0.256 HP)

Mode High Speed — Opsi 3S (Standar)

Motor 2216 1000 KV @3S sudah cukup untuk HS 80 dan 100 km/h — satu motor untuk semua mode.

Parameter	80 km/h	100 km/h
KV dibutuhkan @3S	**~1000 KV**	**~1000 KV**
Motor 1000 KV @3S	**✓ cukup**	**✓ cukup**
Propeller	APC 8×7	**APC 9×8**
Baterai	**3S 10000 mAh**	3S 10000 mAh
ESC	40 A ✓	40 A ✓
Current @3S	~8.5 A	~13.8 A

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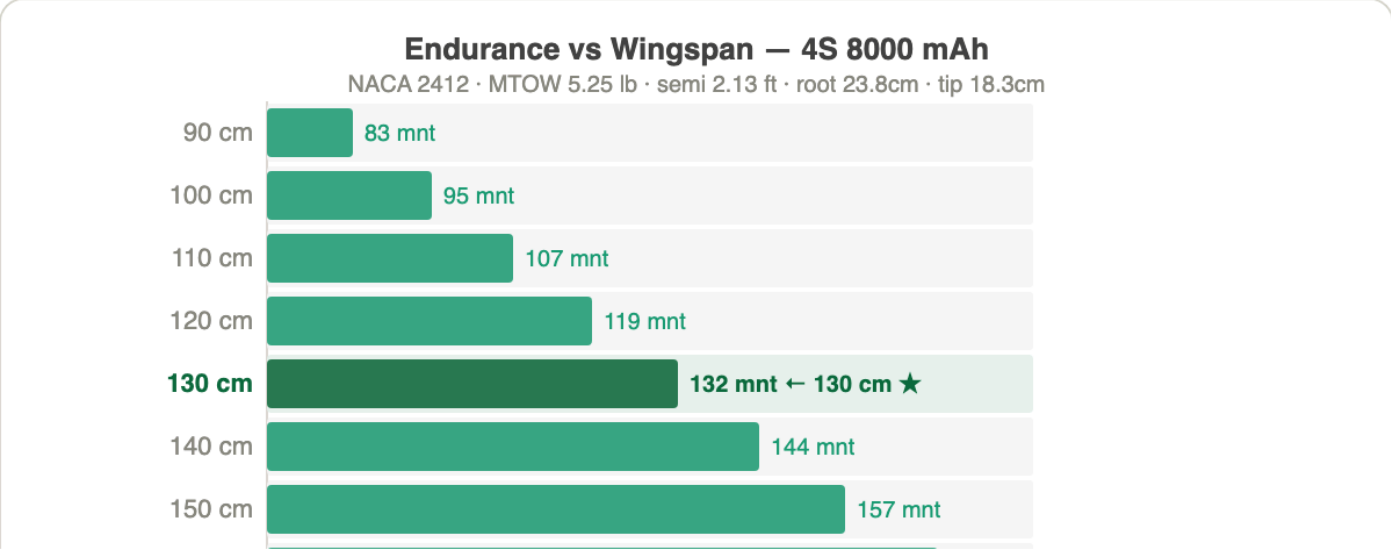
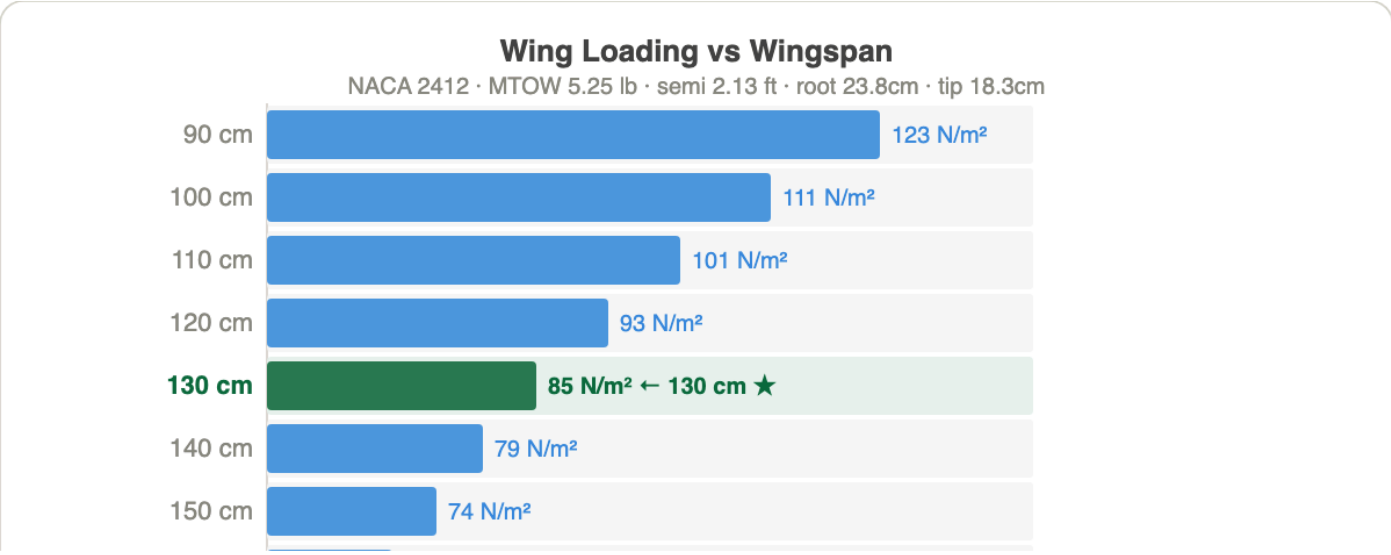
6. Analisis Wingspan

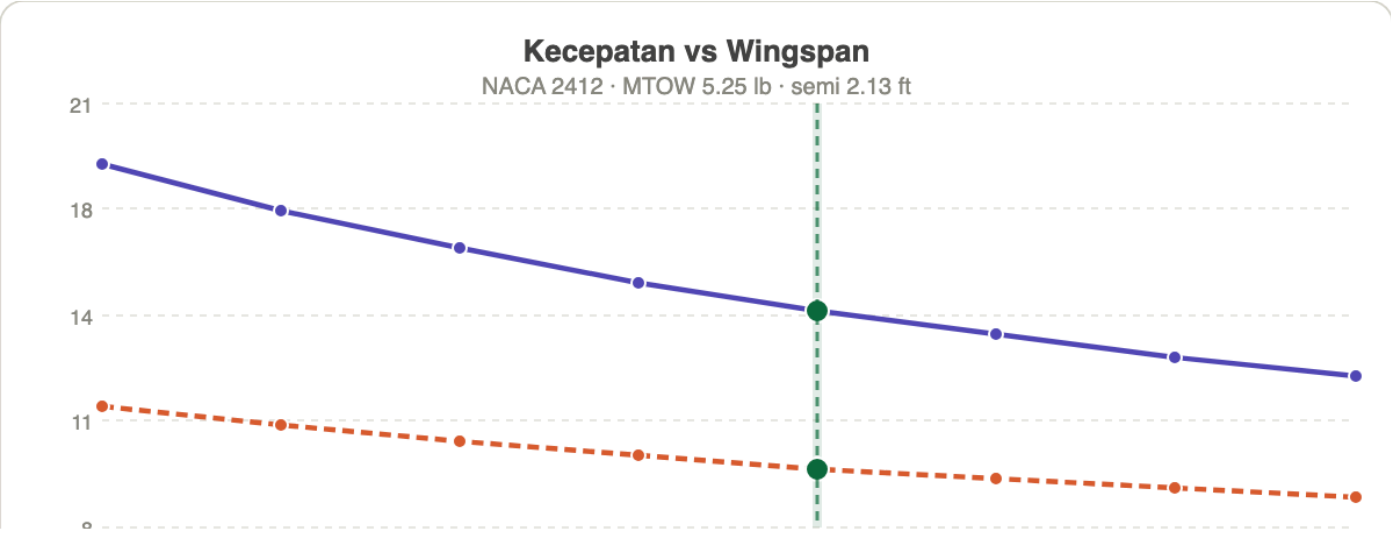
Endurance



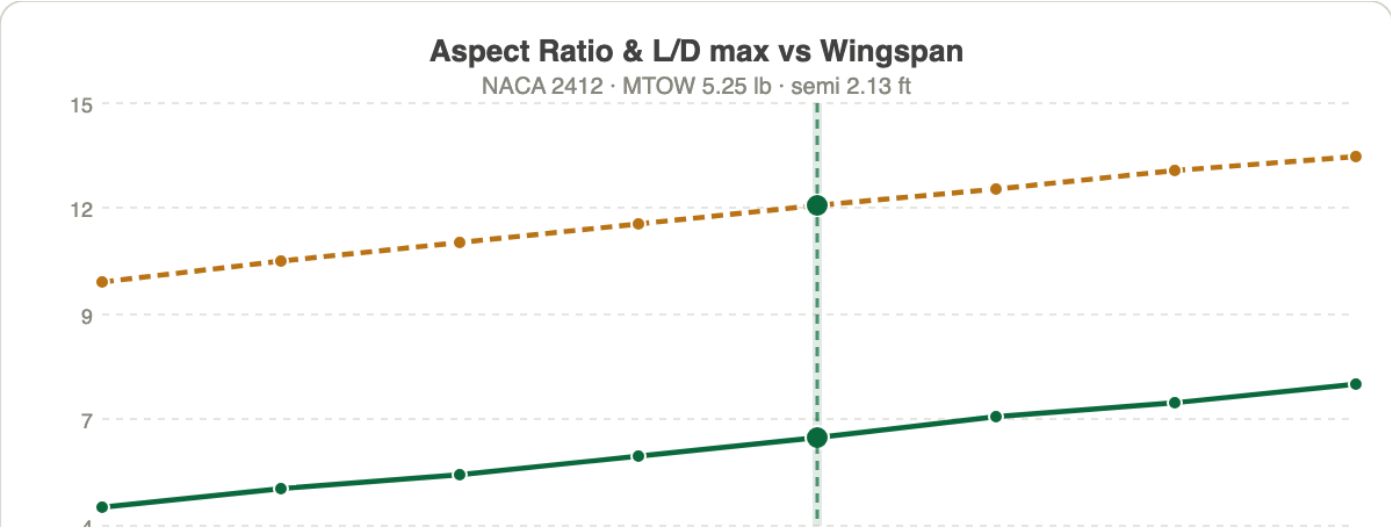
Wing Loading



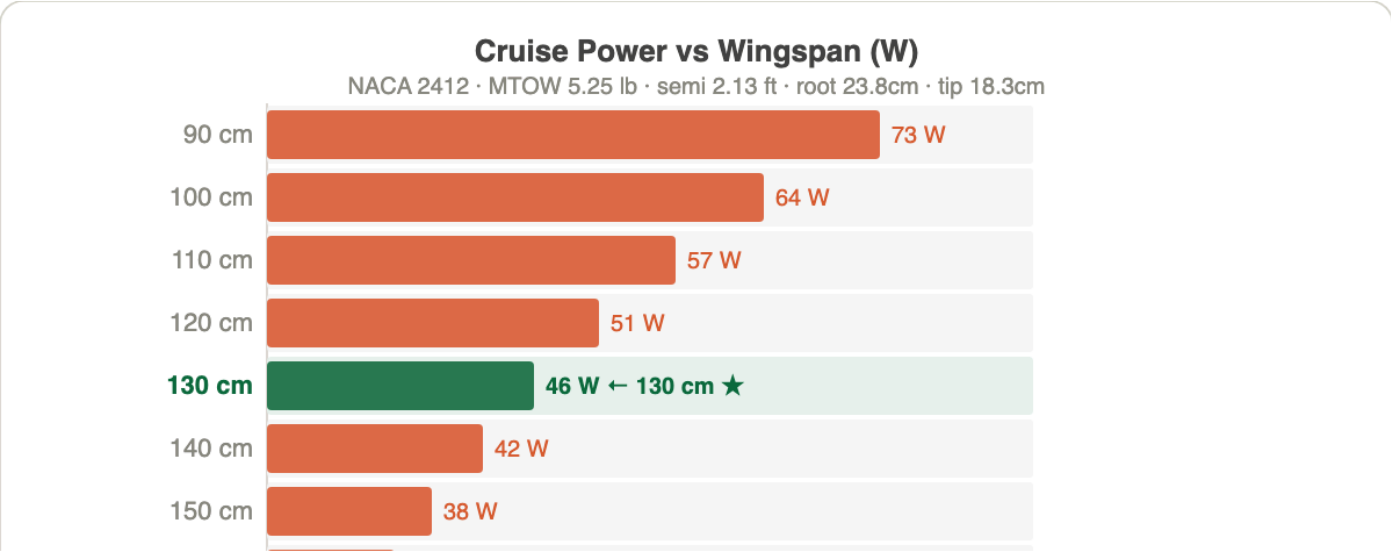
Kecepatan



Aspect Ratio & L/D



Cruise Power

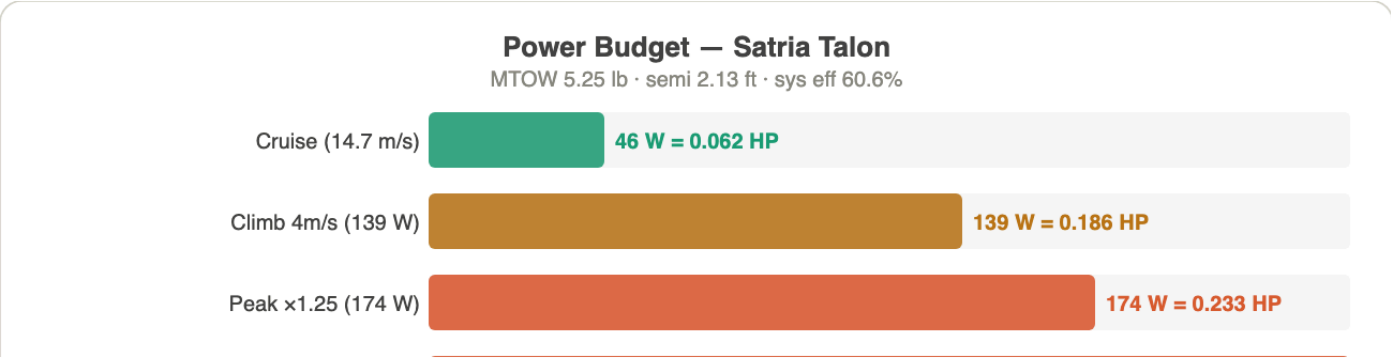


Tabel

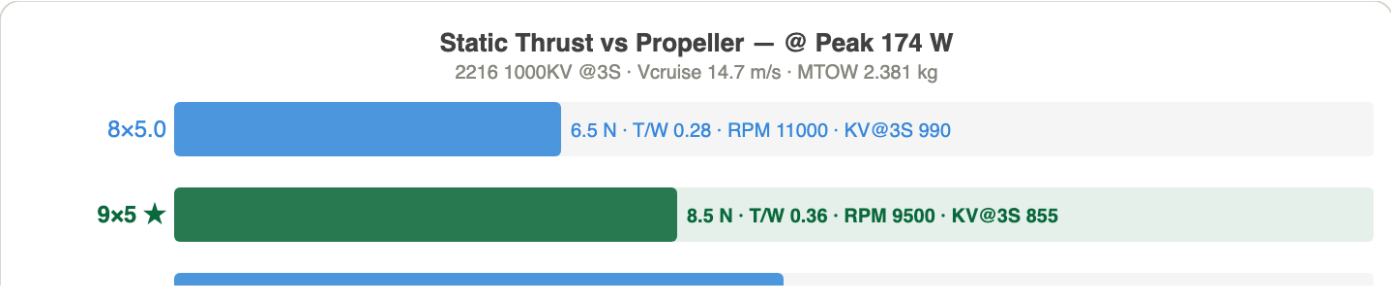
Wingspan	Area	AR	WL	Vs	Vcruise	L/D	P cruise	Endurance (4S 8Ah)
90 cm	0.189 m²	4.3	123 N/m²	11.6 m/s	19.4 m/s	10.3	73 W	83 mnt
100 cm	0.210 m²	4.8	111 N/m²	11.0 m/s	17.9 m/s	10.9	64 W	95 mnt
110 cm	0.232 m²	5.2	101 N/m²	10.5 m/s	16.7 m/s	11.4	57 W	107 mnt
120 cm	0.253 m²	5.7	93 N/m²	10.0 m/s	15.6 m/s	11.9	51 W	119 mnt
130 cm ★	**0.2731 m²**	**6.17**	**86 N/m²**	**9.65 m/s**	**14.74 m/s**	**12.37**	**46 W**	**131 mnt**
140 cm	0.295 m²	6.7	79 N/m²	9.3 m/s	13.9 m/s	12.8	42 W	144 mnt
150 cm	0.316 m²	7.1	74 N/m²	9.0 m/s	13.2 m/s	13.3	38 W	157 mnt
160 cm	0.337 m²	7.6	69 N/m²	8.7 m/s	12.6 m/s	13.7	35 W	171 mnt

7. Propulsi

Power Budget



Propeller



Parameter	Nilai
Motor	**2216, KV 1000 @ 3S**
Power rating	≥ 250 W
Propeller	**APC 9×5**
ESC	**40 A BLHeli_32**
Baterai	**3S 10000 mAh**

Baterai	Endurance	Kargo	C-rate (cruise)	
3S 5000 mAh	~62 mnt	~700 g	0.83C	Cukup
3S 8000 mAh	~99 mnt	~620 g	0.52C	Baik
3S 10000 mAh	**~123 mnt**	**~541 g**	**0.41C**	**✓ Optimal**
4S 8000 mAh	~131 mnt	571 g	0.28C	Alternatif

8. Misi & Jangkauan

Konfigurasi	Kecepatan	Kargo	Endurance	Range one-way	Radius RTH
3S 10000 mAh — cruise	**53.1 km/h**	**~541 g**	**~123 mnt**	**~109 km**	**~44–47 km**
3S 10000 mAh — HS 80	**80 km/h**	**~541 g**	**~60 mnt**	**~80 km**	**~31–34 km**
3S 10000 mAh — HS 100	**100 km/h**	**~541 g**	**~37 mnt**	**~62 km**	**~23–25 km**

Konfigurasi	Kecepatan	Kargo	Endurance	Range one-way	Radius RTH
4S 8000 mAh — cruise	53.1 km/h	571 g	~131 mnt	~116 km	~49–52 km
4S 8000 mAh — HS 80	80 km/h	571 g	~64 mnt	~85 km	~34–36 km
4S 8000 mAh — HS 100	100 km/h	571 g	~39 mnt	~65 km	~25–27 km

9. Rekomendasi Operasional

1. **CG fisik:** First flight di **5.56 cm dari LE** (26.5% MAC, 0.91 ft)
2. **CG nominal: 5.87 cm dari LE** (27.9% MAC, 0.92 ft) setelah first flight validated
3. **Batas keras CG:** Jangan melebihi **7.09 cm dari LE** (33.7% MAC, 0.96 ft)
4. **V-tail mixing:** Konfigurasi ruddervator di ArduPlane/iNav, uji pitch/yaw di darat sebelum terbang
5. **Pitot tube:** Wajib — Vs 9.65 m/s, airspeed measurement kritis
6. **Baterai:** 3S 10000 mAh untuk endurance optimal; C-rate 0.41C sangat aman

10. Ringkasan Spesifikasi Final

Parameter	Nilai
Airfoil	**NACA 2412**
Semi-length	**2.13 ft (64.92 cm)**
Wingspan	**129.84 cm**
Root / Tip chord	**23.77 cm / 18.29 cm**
Taper ratio	0.7692
Sweep	0.0°
Wing area	0.2731 m²
Aspect ratio	6.1739
MAC	21.03 cm @ ±31.0 cm dari CL
Konfigurasi ekor	V-tail twin fin (45°)
Empty weight	**3.00 lb = 1.361 kg**
MTOW	**5.25 lb = 2.381 kg**
Payload total	1.021 kg
CG fwd	**0.91 ft = 5.56 cm dari LE = 26.5% MAC**
CG nominal ✓	**0.92 ft = 5.87 cm dari LE = 27.9% MAC**
CG aft limit	**0.96 ft = 7.09 cm dari LE = 33.7% MAC**
Wing loading	85.5 N/m²
Stall speed	**9.649 m/s (34.7 km/h)**
Cruise speed	**14.738 m/s (53.1 km/h)**
VNE	33.77 m/s
L/D max	12.366
Cruise power	45.9 W (0.0616 HP)
Peak power	174 W (0.234 HP)

Parameter	Nilai
Endurance cruise (3S 10Ah)	**~123 mnt**
Range one-way (cruise)	**~109 km**
Radius RTH (cruise)	**~44–47 km**
Cruise power HS 80	94.0 W (0.126 HP)
Endurance HS 80 (3S 10Ah)	**~60 mnt**
Range one-way HS 80	**~80 km**
Radius RTH HS 80	**~31–34 km**
Cruise power HS 100	153 W (0.205 HP)
Endurance HS 100 (3S 10Ah)	**~37 mnt**
Range one-way HS 100	~62 km
Radius RTH HS 100	~23–25 km
Motor	**2216, KV 1000 @ 3S**
Propeller (cruise)	APC 9×5
Propeller HS 80	APC 8×7
Propeller HS 100	**APC 9×8**
ESC	40 A BLHeli_32
Baterai	**3S 10000 mAh**
Baterai (4S opsi)	4S 8000 mAh

UAV Satria Talon · NACA 2412 · Semi 2.13 ft · Root 0.78 ft · Tip 0.60 ft · Sweep 0° · V-tail
Empty 3.00 lb · MTOW 5.25 lb · Seluruh nilai performa adalah estimasi teoritis.